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A measure of absorptive capacity: Scale development and validation ☆

Tessa C. Flatten [a,*], Andreas Engelen [a], Shaker A. Zahra [b], Malte Brettel [a]

[a] *Aachen University (RWTH), Templergraben 64, 52056 Aachen, Germany*
[b] *Department of Strategy & Organization, Gary S. Holmes Center for Entrepreneurship, Carlson School of Management, University of Minnesota, 321 19th Avenue South, Minneapolis, MN 55455, USA*

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Summary Academic interest in absorptive capacity (ACAP), which has grown rapidly over the past two decades, has focused on ACAP's effect on organizational learning, knowledge sharing, innovation, capability building, and firm performance. Even though Cohen and Levinthal's work (1990) highlights the multidimensionality of ACAP, researchers have measured it as a uni-dimensional construct, often using a firm's R&D spending intensity as a proxy for this construct. This practice raises questions about the veracity of the claims made in the literature about the nature and contributions of ACAP. The present study develops and validates a multidimensional measure of ACAP, building on relevant prior literature, a series of pre-tests, and two large survey-based studies of German companies.
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Introduction

Over the last two decades, the concept of absorptive capacity (ACAP) has received considerable attention in the literature. Building on the work of Cohen and Levinthal (1989, 1990), researchers have shown that ACAP influences innovation (Tsai, 2001), business performance, intraorganizational transfer of knowledge (Gupta & Govindarajan, 2000; Szulanski, 1996), and interorganizational learning (Lane & Lubatkin, 1998; Lane, Salk, & Lyles, 2001). Cohen and Levinthal (1989) conceptualize ACAP as the firm's ability to identify, assimilate, and exploit knowledge gained from external sources. As such, ACAP facilitates knowledge accumulation and its subsequent use. Because exploiting externally acquired knowledge usually requires converting its content into a usable form, Zahra and George (2002) broaden ACAP from the original three dimensions (identify, assimilate, and exploit) to four dimensions (acquire, assimilate, transform, and exploit).

Even though a considerable number of empirical studies have used ACAP, a valid measure that incorporates its various dimensions has not yet been developed (Wang & Ahmed, 2007). Lane, Koka, and Pathak (2006) observe that most researchers typically measure ACAP with simple R&D proxies

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* Corresponding author. Tel.: +49 241 80 96197; fax: +49 241 80 92177.
E-mail address: flatten@win.rwth-aachen.de (T.C. Flatten).

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(e.g., Cohen & Levinthal, 1989), ignoring the variety of its dimensions and their implications for different organizational outcomes. Table 1, which provides an overview of the various proxies employed to capture ACAP in prior studies, shows that researchers have used as proxies a firm's R&D outputs (e.g., patents) and inputs, such as R&D intensity (which Cohen & Levinthal, 1990) defined as R&D expenditures divided by sales) and investments (R&D personnel).

The use of these proxies may have contributed to conflicting and misleading findings about the nature and contributions of ACAP. For example, companies differ in their propensity to patent their innovations, so the use of patents may understate the firm's ACAP. Patents also differ greatly in terms of their knowledge content (Coombs & Bierly, 2006), so it is unclear whether the use of patents fully captures ACAP. Likewise, R&D spending is not the only source of ACAP since employee skills, organizational memory, and prior organizational experiments and experiences contribute significantly to a firm's overall ACAP. Further, because ACAP is potentially a multidimensional construct (Zahra & George, 2002), it is debatable whether any single dimensional measure can fully gauge this complex construct. These concerns lead Lane et al. (2006, p. 858) to suggest that ''absorptive capacity should be empirically explored in non-R&D contexts using metrics that capture each dimension of the absorptive capacity process in a manner appropriate for that context.'' Lane and colleagues also note that studies that use proxies to measure ACAP cannot capture the complexity of its various dimensions because such measures usually treat ''absorptive capacity as a static resource and not as a process or capability'' (Lane et al., 2006, p. 838). These shortcomings suggest a need for a more valid measure that captures the multiple dimensions of ACAP.

Other existing operationalizations of ACAP (e.g., Jansen, Van den Bosch, & Volberda, 2005; Szulanski, 1996; Thérin, 2007) have weaknesses that compromise their validity. For example, Szulanski (1996) research is restricted to eight companies, possibly too small for the results to be generalizable, and the scale development process was conducted

Table 1 Studies using proxies to measure ACAP.

Author	Research topic	Proxy
Ahuja and Katila (2001)	Technological acquisition and firm performance	Number of patents
Belderbos, Carree, Diederen, Lokshin, and Veugelers (2004)	Heterogeneity in R&D co operations	R&D-intensity
Boynton, Zmud, and Jacobs (1994)	Influence of IT management processes concerning the IT usage in large companies	IT knowledge of the management
Cockburn and Henderson (1998)	R&D productivity	Number of academic publications
Cohen and Levinthal (1989)	R&D investments	R&D-intensity
Lenox and King (2004)	ACAP development on management level	Knowledge management (flow of information)
Liu and White (1997)	Investments in the Chinese industry	Investments in R&D employees
Meeus, Oerlemans and Hage (2001)	Learning	R&D-intensity
Mowery et al. (1996)	Strategic alliances and in-house knowledge transfer	Patents and R&D-intensity
Mowery and Oxley (1995)	Technological transfer and national innovation ability	Investments in technical and academic further education
Mukherjee, Mitchell, and Talbot (2000)	Success of product plant	Labor productivity and compliancy quality
Muscio (2007)	Effects of co operations in SME	In-house items: degree of employees which are assigned with R&D activities or in-house education
Nielsen and Pawlik (2007)	Export intensity of foreign affiliates	Wage-level of foreign companies compared to the level of domestic companies
Oltra and Flor (2003)	Innovation output of a company	R&D-intensity
Stock, Greis, and Fischer (2001)	New product development	R&D-intensity
Tsai (2001)	Firm performance and innovation success	R&D-intensity
Vandenbosch, Volberda, and De Boer (1999)	Organizational form and ability	Incentive system
Veugelers (1997)	Level of innovation activities	Employee of R&D, postgraduates in R&D, proportion of R&D in basic research
Vinding (2006)	Innovation success	HR management

with overlapping samples; both adequate sample size and two non-overlapping samples are requirements for a valid and reliable scale (Churchill, 1979; DeVellis, 2003). Similarly, the generalizability of Thérin (2007) scale is limited because the study was conducted with only one sample that focused on entrepreneurial firms. Jansen et al. (2005) study first used a survey of 15 CEOs, but the scale development was conducted with managers from one company in the financial services industry, so the small size of the sample limits the generalizability of the results.

This paper reports on an empirical study that develops a measure of ACAP that can be valuable extension to the simple proxies that are commonly used in the literature and assesses its psychometric properties. This measure can help address the question of whether ACAP has three dimensions, as proposed by Cohen and Levinthal (1990), or four, as suggested by Zahra and George (2002). Following the established procedures for scale development (e.g., Churchill, 1979; DeVellis, 2003) and combining qualitative and quantitative methods achieve these objectives. We chose a combination of qualitative and quantitative methods in order to profit from their combined benefits (a ''mixed-method approach''). Qualitative methods provide important insights into under-researched phenomena, but they are prone to subjectivity and lack generalizability. Quantitative methods achieve higher degrees of generalization, mostly by means of statistical tools (Hurmerinta-Peltomäki & Nummela 2006; Jick 1979; Mohr 1982). In applying this mixed-method approach, we used a sequential research design. Based on DeVellis (2003), we first conducted qualitative analyses by performing a literature review, keyword analyses and interviews in order to generate an initial item pool. Subsequently, we applied quantitative methods to test the resulting item pool for reliability and validity in our two surveys (Churchill, 1979).

The following sections discuss and clarify the ACAP construct, present the items generated based on the literature and describe the process used to purify this pool. A summary of several rounds of questionnaire pre-testing and a description of the two large samples used in the research follow (qualitative methods). Next, the quantitative methods follow: The first sample tests several theoretically plausible alternative factor structure specifications and eliminates those items that do not adequately reflect the theoretical components of the ACAP construct. The second sample is used to replicate the findings of the first sample. The final section of the paper discusses the results and identifies promising avenues for further research.

The construct of absorptive capacity

ACAP refers to a firm's ability to recognize the value of new external knowledge, assimilate it, and apply it to commercial ends. Key antecedents of ACAP include prior related knowledge (which usually includes basic skills and experience) and organizational factors such as the structure of communication and distribution of knowledge (Allen, 1983; Evenson & Kislev, 1975; Tilton, 1971). Cohen and Levinthal's (1989, 1990) original definition highlights three dimensions of ACAP–knowledge identification, assimilation, and exploitation to a commercial end–but several re-conceptualizations of the original ACAP construct have appeared in the literature (e.g., Jansen et al., 2005; Torodova & Durisin, 2007; Van Den Bosch, Van Wijk, & Volberda, 2003; Zahra & George, 2002). In line with recent research (Jansen et al., 2005; Liao, Welsch, & Stoica, 2003; Tu, Vonderembse, Ragu-Nathan, & Sharkey, 2006), this study follows the re-conceptualization offered by Zahra and George (2002), who distinguish between potential ACAP (knowledge acquisition and assimilation) and realized ACAP (knowledge transformation and exploitation).

Acquisition refers to a firm's ability to identify and obtain knowledge from external sources (e.g., suppliers). Assimilation refers to a firm's ability to develop processes and routines useful in analyzing, interpreting, and understanding externally acquired knowledge (Szulanski, 1996). Transformation means developing and refining those routines that facilitate combining existing knowledge with acquired and assimilated knowledge for future use (Zahra & George, 2002). Exploitation denotes a firm's capacity to improve, expand, and use its existing routines, competencies, and technologies to create something new based on the ''transformed'' knowledge (del Carmen Haro-Dominguez, Arias-Aranda, Javier Lloréns-Montes, & Ruiz Moreno, 2007).

Together, the four dimensions of ACAP enable companies to exploit new discoveries and knowledge (Cohen & Levinthal, 1994) and serve as a crucial, intangible resource that can enhance firm performance (Barney, 1991; Wernerfelt, 1984) and be a major competitive advantage (Teece, Pisano, & Shuen, 1997). ACAP achieves competitive advantage primarily through innovation and strategic flexibility (Zahra & George, 2002). The steps of potential ACAP lead to ''renewing a firm's knowledge base and the skills necessary to compete in changing markets'' (Zahra & George, 2002, p. 196); therefore, ''firms that are flexible in using their resources and capabilities can reconfigure their resource bases to capitalize upon emerging strategic opportunities'' (Zahra & George, 2002, p. 196). Superior performance follows from their first-mover advantages and responsiveness to customers. Realized ACAP consists of transformation capabilities, which enable firms to develop new processes or to add changes to existing processes, and exploitation capabilities, which are used to convert knowledge into new products to enhance performance and competitive advantage.

Having described the domain of ACAP and explained its dimensions, an empirical study that attempts to capture these dimensions follows, beginning with identifying items that could be used to gauge ACAP.

Scale item generation

In line with Newell and Goldsmith (2001), a measure of ACAP was developed by closely following the established process of item generation and scale development (Churchill, 1979; DeVellis, 2003). A literature review was conducted by screening all articles published in ten Management Journals (*Academy of Management Journal, Academy of Management Review, Administrative Science Quarterly, Journal of Management, Journal of Management Studies, Management Science, Organization Science, Strategic*

Management Journal, *MIS Quarterly*, and *European Management Journal*) from 1990 to 2007 to identify related research streams that are similar to or overlap with at least one dimension of ACAP. We chose these journals based on two criteria: the journals had to have a strong focus on general management and organization topics so they were likely to cover ACAP-related studies. Second, after leaving out methodological and niche-topic journals (like journals on information technology), the journals had to be ranked in the top positions in the management category in the 2008 ISI Journal Citation Report. For the second criteria, we also checked consistency with other prominent journal rankings–such as the study from Tahai and Meyer (1999), the 2006 ELJ ranking, the 2008 VHB ranking, and the ranking from the 2006 Financial Times Survey of Top Business Schools–to ensure that these rankings also placed these general management journals in their top positions. As a result of this process, the set of journals we chose gave us reason to believe that we were identifying the ACAP research with the strongest impact. The timeframe was chosen based on the fact that the ACAP research stream was introduced by Cohen and Levinthal in September 1989, and our first interviews were conducted in January 2008.

In addition to studying all articles published in these journals in terms of their relevance to ACAP, we researched databases to identify relevant studies not published in these journals. The review of Lane et al. (2006) identifies four research streams with strong overlap with ACAP: organizational learning, strategic alliances, knowledge management, and the resource-based view. Drawing upon these four research areas, we searched for relevant studies in electronic databases, such as Science Direct, JSTOR, and Business Source Premier, using keywords of these research streams to find relevant studies that were not included in the initial set of ten journals. Using the subject thesaurus term function in these databases, we cross-checked to ensure that we missed no similar subjects and identified related research streams (Feldvari, 2000; Knapp, 2000). To avoid focusing only on published studies, we also included databases like the Social Sciences Research Network (SSRN) and conference papers. All together, we identified 269 studies from 29 related research areas that were either theoretical, qualitative or survey-based. As Figure 1 shows, most of the studies were theoretical or qualitative and contained no measurement instruments.

An important preliminary step in the development of an initial item pool was the matching of related research streams to a particular ACAP dimension. Therefore, we compared the characteristics derived from the most prominent definitions of each research area with the characteristics of each ACAP dimension in order to assign the research streams to one or more ACAP dimensions, as depicted in Table 2. In other words, Table 2 assigns items from specific related research streams to the dimensions of the ACAP construct; for example, Table 2 shows that items derived from the literature on environmental scanning are related to the dimension of acquisition and belong in the respective initial item pool. This step is necessary since it is our intention to develop a multi-dimensional measurement scale of ACAP that addresses the major weakness the proxies that have been used for ACAP by extant research. Once items were assigned to the ACAP dimensions, if measurement scales were available for the particular theory, we followed Hinkin (1995) in using those items as an initial item pool for the ACAP scale development process.

As such, while being targeted at broad types of information, acquisition, as a dimension of ACAP, shares with market-oriented intelligence generation the generation of

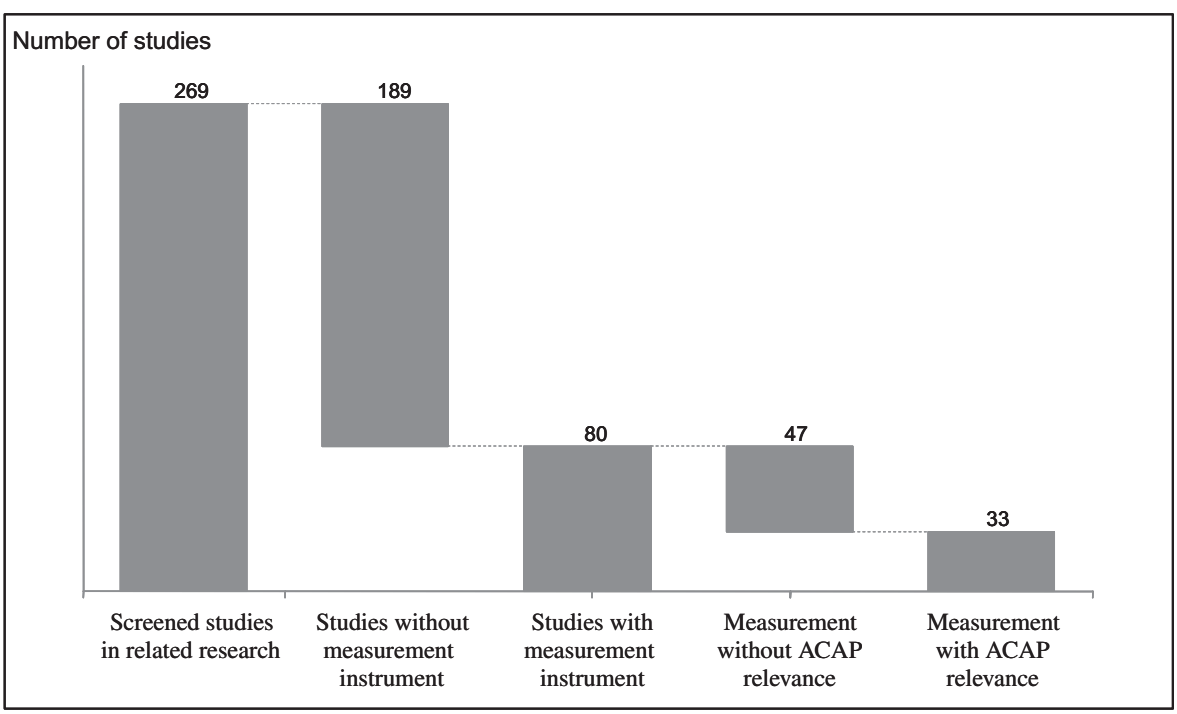

Figure 1 Studies screened to generate the initial item-pool.

Table 2 Overlaps and similarities of ACAP's dimensions with related research streams.

Related research stream	Definition of related research stream	Domain of overlap and similarities to ACAP (on level of dimensions)
Collective mind	''Collective mind is conceptualized as a pattern of heedful interrelations of actions in a social system. Actors in the system construct their actions (contributions), understanding that the system consists of connected actions by themselves and others (representation), and interrelate their actions within the system (subordination)'' (Weick & Roberts, 1993, p. 38)	*Assimilate*: — Communication of the knowledge and a kind of shared language and understanding
Environmental scanning	''Scanning (...) includes doing a broad sweep of the horizon to look for signs of change and opportunities (...). Scanning activities could range from gathering data deliberately such as by doing market research, to informal conversations with other executives, or reading the newspaper'' (Auster & Choo, 1993, p. 44)	*Acquire*: — Identification of knowledge in various external sources by a broad range of activities (e.g., formal and informal means)
Exploration	''Firms strive to develop capabilities to excel at the creation or acquisition of new knowledge'' (Bierly & Daly, 2007, p. 494)	*Acquire*: — Identification of new knowledge in various external sources *Transform:* — Creation of new knowledge
Idea generation	''A systematic search for new product ideas'' (Kotler & Armstrong, 1991, p. 283)	*Transform:* — Open-mindedness to new ideas and insights
Information processing	''Gathering, interpreting, and synthesis of information in the context of organizational decision making'' (Tushman & Nadler, 1978, p. 283)	*Acquire:* — Identification of knowledge in various external sources *Assimilate*: — Shared interpretation of the acquired knowledge
Information search	The degree of attention, perception, and effort directed toward obtaining environmental data or information related to the specific purchase under consideration (Beatty & Smith, 1987)	*Acquire:* — Major importance of information generation, particularly information on environmental data that is relevant to the current problem
Innovation capability	''The ability to continuously transform knowledge and ideas into new products, processes and systems for the benefit of the firm and its stakeholders'' (Lawson & Samson, 2001, p. 384)	*Transform:* — Continuous transformation of knowledge, e.g., into new product ideas *Exploit:* — Use of transformed knowledge for product development and for the benefit of the overall organization
Innovation management	Five activities together define innovation management: technological integration, new product development process, strategic technology planning, organizational change, and business development (Drejer, 2002)	*Transformation:* — Combination of knowledge to push the company's technological success
Interorganizational learning	''Interorganizational learning (IOL) (...) consists of three sub-constructs: information sharing, the development of relational memory, and shared meaning for mutual understanding'' (Choi & Ko, 2010, p. 1)	*Assimilate:* — Communication of knowledge and integration of this knowledge into the organizational knowledge base
Knowledge acquisition	Knowledge acquisition is defined as the development or creation of skills, insights and relationships (DiBella & Nevis, 1998)	*Acquire*: — Generation of insights from various sources *Assimilate*: — Basic conversion of the acquired knowledge
Knowledge creation	''The capability of a company as a whole to create new knowledge, disseminate it throughout the organization, and embody it in products, services, and systems'' (Nonaka & Takeuchi, 1995, p. 3)	*Assimilate*: — Dissemination of knowledge throughout the organization *Transform*: — Conversion and combination of knowledge *Exploit*: — Commercial use of knowledge, e.g., for the development of new products or technologies

(*continued on next page*)

Table 2 (*continued*)

Related research stream	Definition of related research stream	Domain of overlap and similarities to ACAP (on level of dimensions)
Knowledge dissemination	Knowledge dissemination is defined as sharing what has been learned. It is the process by which knowledge is shared and diffused throughout the organisation (Argyris & Schön, 1978)	*Assimilate:* — Dissemination of knowledge throughout the organization
Knowledge exchange	''The smoothness of knowledge transaction and the outcome of exchanged knowledge in terms of quality and quantity in the internal knowledge market''(Fang, Huang, & Liu, in press, p. 18)	*Assimilate*: — Dissemination of knowledge throughout the organization
Knowledge exploitation	Use and further development of existing competencies (March, 1991) and ''learning activities involving the use of resources the firm already has'' (Liu, 2006, p. 145)	*Transform:* — Use of existing resources (such as knowledge) to create new resources *Exploit:* — Commercial use of existing knowledge within the organization, e.g., to develop new products or technologies
Knowledge generation	''A knowledge manipulation activity that produces knowledge by processing existing knowledge where the latter has been acquired by selection, acquisition and/or prior generation'' (Holsapple & Joshi, 2002, p. 14)	*Transform*: — Creation of new knowledge based upon existing knowledge bases within the organization
Knowledge identification	The focus is on checking the availability of knowledge in one's mind deemed necessary for effectively coping with the affordances of a particular cognitive task, e.g., attaining a particular instructional goal, solving a complex problem, acquiring expert knowledge (Tergan, 2003)	*Acquire*: — Identification of knowledge out of various sources
Knowledge integration	The process of absorbing knowledge from external sources and blending it with the technical and business skills, know-how, and expertise that reside in the business and IS units of a firm (Grant, 1996)	*Acquire:* — Generation of knowledge out of various external sources *Transform:* — Combination of existing knowledge and newly generated knowledge
Knowledge management	''Systemic and organizationally specified process for acquiring, organizing, and communicating both tacit and explicit knowledge...'' (Alavi & Leidner, 1999, p. 2)	*Acquire:* — Acquisition of knowledge from various sources *Assimilate:* — Communication of various types of knowledge (e.g., tacit and explicit knowledge)
Knowledge sharing	Knowledge sharing requires the dissemination of individual employees ' work-related experiences and collaboration between and among individuals, subsystems, and organizations (Dyer, 1997)	*Assimilate*: — Dissemination of knowledge of employees throughout the organization
Knowledge transfer	''Incorporates the processes that shape both the collection and dissemination of knowledge within the organization'' (Watson & Hewett, 2006, p. 142)	*Acquire*: — Identification and generation of knowledge out of various sources *Assimilate*: — Dissemination of knowledge within the organization
Knowledge distribution	''We define the distribution of knowledge in terms of who has what type of information'' (Rulke & Galaskiewicz, 2000, p. 613)	*Assimilate:* — Assortment of relevant information
Learning capacity	The potential level of organizational knowledge diffusion over a given time period based upon set conditions (Schreiber & Carley, 2008)	*Assimilate*: — Dissemination of knowledge within the organization
Market orientation: Intelligence dissemination	The process and extent of knowledge exchange within a given organization (Kohli et al., 1993)	*Assimilate*: — Dissemination of knowledge within the organization

Table 2 (*continued*)

Related research stream	Definition of related research stream	Domain of overlap and similarities to ACAP (on level of dimensions)
Market orientation: Intelligence generation	Collection and assessment of both customer needs/preferences and the forces that influence the development and refinement of those needs (Kohli et al., 1993)	*Acquire*: – Generation of knowledge from external sources, particularly knowledge on market conditions
Market orientation: Responsiveness	Action taken in response to intelligence that is generated and disseminated (Kohli et al., 1993)	*Exploit*: – Use of generated and disseminated knowledge in market activities (e.g., product development)
Organizational learning	''Process through which an organization expands its repertoire of actions, and they focus on how knowledge is acquired and distributed'' (Edmondson & Moingeon, 1998, p. 24)	*Acquire:* – Identification of knowledge out of external sources *Assimilate:* – Communication, discussion of the knowledge *Transform:* – Creation of new resources or knowledge
Organizational memory	''In the organizational view, organization members' actions may lead to organizational interactions with the world, which results in outcomes that are interpreted by people and shared among members, creating organizational memory in the form of shared beliefs, values, assumptions, norms, and behaviors'' (Moorman & Miner, 1998, p. 92)	*Assimilate*: – Creation of shared information and knowledge within the organization
Team knowledge	The collection of task- and team-related knowledge held by teammates and their collective understanding of the current situation (Cooke, Salas, Cannon-Bowers, & Stout, 2000)	*Assimilate:* – Communication and dissemination of knowledge to achieve collective understanding

market-related knowledge (Kohli, Jaworski, & Kumar, 1993), with environmental scanning (Auster & Choo, 1993) and knowledge management (Alavi & Leidner, 1999) an alertness to new and relevant information outside the organization, and with the research stream of ''information search'' how an organization permanently invests in obtaining environmental data (Beatty & Smith, 1987). The acquisition dimension is also related to research on information processing, knowledge acquisition, knowledge exploration, knowledge identification, knowledge integration, knowledge transfers, organizational learning, and team knowledge. These overlaps and similarities with these research streams are shown in Table 2. Thus, items already used in these research streams can serve as a starting point for an ACAP scale development process.

As for assimilation, information processing (Tushman & Nadler, 1978) shares with this ACAP dimension a focus on the shared interpretation of acquired knowledge, while knowledge dissemination (Argyris & Schön, 1978) and organizational learning (Edmondson & Moingeon, 1998) share with assimilation the dissemination of knowledge throughout the organization, and interorganizational learning shares the component of knowledge communication within the organization (Choi & Ko, 2010). As outlined in Table 2, based upon prominent definitions of these related research streams, the assimilation dimension also relates to research on the collective mind, intelligence dissemination, knowledge creation, management, acquisition, exchange, exploration, management sharing and transfer, learning capacity, organizational memory and team knowledge.

Transformation also shares overlaps and similarities with some related research streams. Like transformation, knowledge exploitation is characterized by the development of existing resources; knowledge generation describes the combination of existing knowledge to create new knowlege which shows clear overlaps with transformation (Holsapple & Joshi, 2000); knowledge integration focuses on blending new knowlege with existing resourcs (Grant, 1996); and knowlege creation deals with how new knowledge can be generated within an organization and thereby captures a major facet of transformations as ACAP dimension (Nonaka & Takeuchi, 1995). The transformation dimension also relates to research on idea generation, innovation capability, innovation management, and organizational learning (Table 2).

As noted earlier, extant research has paid little attention to the exploitation of knowledge, suggesting a need for the development of items that view transformation as a strategy with which organizations can increase their intellectual capital by creating unique knowledge and useful commercial applications (Choo & Bontis, 2002). Research streams related to exploitation are limited to those of innovation capability, market-oriented responsiveness, knowledge creation and knowledge exploitation. Exploitation and knowledge creation, for example, have in common the goal of the commercial use of knowledge, such as in new products (Nonaka & Takeuchi, 1995).

As Figure 1 shows, 33 of the 269 papers in the related research streams contain items that pertain to at least one dimension of ACAP and that could be useful in developing the ACAP scale. The other 236 studies were either of a theoretical or qualitative nature or did not provide any items that were relevant to any ACAP dimension. From these 33 studies we chose an initial item pool of 52 items for their relevance, uniqueness, and ability to convey to informants ''different shades of meaning'' of the ACAP construct (Churchill, 1979, p. 68). In the end, twelve items described the dimension *acquire*, fifteen the dimension *assimilate*, nineteen the dimension *transform*, and six the dimension *exploit*.

Our initial pool of items focuses on activities that form the four dimensions of ACAP, but proxies that the extant literature on ACAP has frequently employed (such as R&D spending; compare Table 1) were not considered for two primary reasons: First, the use of a single static proxy is not in line with the dynamic nature of the ACAP construct, which is better captured by activities that reflect dynamic processes (Lane et al., 2006). Second, these proxies do not relate to the ''core'' of the ACAP construct but to its antecedents and/or consequences. For example, the proxy ''number of patents'' is an outcome of the activities of the ACAP construct, rather than an integral part of these activities.

Pre-testing

Next, we conducted three pre-tests to assess the quality of the 52 items. In pre-test 1, a brief questionnaire containing the items was given in person to 10 executives, who were asked to point out any items that were either ambiguous or difficult to answer; their feedback resulted in the elimination of two items and the addition of three. This process expanded the items to 53, all of which used seven-point Likert-type response scales.

After completing the pre-test 1, we fielded pre-test 2 to 11 academic experts. Respondents provided detailed comments that led to the modification of some and the elimination of other items, resulting in a scale of 36 items (Table 3). The paucity of research on the exploitation dimension meant relying heavily on expert interviews in generating relevant items.

Finally, pre-test 3 of the 36 items with five executives, who were asked to fill out the questionnaire and identify any problems they encountered when completing the scales, resulted in the identification of only a few concerns, prompting minor refinements in the items and instructions. Overall, the results of the three pre-tests suggested that the items intended to gauge ACAP were developed to the point at which a full-scale test was necessary.

Figure 2 summarizes the steps undertaken in the pre-testing phase.

Data collection

Following the pre-tests, the survey was mailed to two large samples drawn from German companies. Two samples were necessary because a second independent sample applied the ACAP measure that was developed based on the first sample, establishing the generalizability of the ACAP measure (Hinkin, 1995). For both samples, companies were randomly drawn from the membership data of the German Chamber for Industry and Commerce. Companies of different sizes and ages were targeted, focusing on research-intensive sectors of the German economy, where ACAP is especially important (Burgel & Murray, 2000): the chemical, mechanical, and electrical engineering industries (Standard Industrial Classification 28–38). In each case, the CEO of each company received a personalized e-mail because these executives were considered to be the most knowledgeable about their companies' operations (Vanderwerf & Brush, 1989).

The target population for the first sample consisted of 2497 executives. The final response calculation excluded 228 of these (e.g., no longer with the firm), leaving a base of 2269. Of these, 285 responded for a final response rate of 13%. The target population for the second sample consisted of 3844 executives, with 311 exclusions from the final response calculation (e.g., executives no longer with the firm), leaving a base of 3533. Of these, 361 responded, for a final response rate of 10%. The two response rates are typical for web-based surveys (Klassen & Jacobs, 2001). The first 2497 firms were contacted between April and June 2008, and the remaining 3844 were contacted between September and November 2008. Two reminders were sent in both cases to increase the response rate. At the time of the survey, approximately half of the firms were less than 50 years old, while the remaining companies were between 51 and 150 years old. About 65% of responding companies had fewer than 150 full-time employees, and the remaining 35% employed between 151 and 700 people.

To test the samples for informant bias, the completed questionnaires were classified, based on hierarchical levels, into two groups: CEOs vs. employees. (Respondents had to indicate whether they were CEO, senior executive or employee). For the first sample's 283 questionnaires, 204 were answered by CEOs, 59 by senior executives and 14 by employees. The latter two groups were merged to one ''non-CEO'' group. For the second sample's 361 questionnaires, 277 were completed by CEOs, 71 by senior executives and two by employees. Again, the latter two groups were merged. Next, the CEO and non-CEO groups were compared for significant differences on the item mean level on a .05 level for both samples, with the result that the low number of significant differences in their replies indicated that there was no bias. Both samples were also tested for non-response bias by comparing early and late respondents (Armstrong & Overton, 1977), with the result that there were no significant statistical differences, indicating that the sample represented its population.

Further, because the data for the dependent and independent variables were collected from a single informant employing a single survey instrument, a test for common method bias was required (Organ & Greene, 1981), and the single-factor test to establish the presence of this bias applied (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). Principal components factor analyses showed that multiple factors emerged, but no single factor accounted for the majority of the co-variance in the measures. For the first sample, 11 factors occurred and the explained variance was 30.4%, and for the second sample, 24 factors emerged

Table 3 Item-pool and sources after three rounds of pre-tests.

Item No.	Item	Source (items are based on)
Acquire 1	Our management emphasizes the exchange of information and experience with companies within the same industry.	Auster and Choo (1993), Veugelers and Cassiman (1999), Wilkens et al. (2004)
Acquire 2	Our management engages in joint research projects with companies and research institutions beyond the industry.	Jansen et al. (2005), Laursen and Salter (2006)
Acquire 3	A periodical meeting with external experts within our industry for the accumulation of relevant information goes without saying in our company.	Auster and Choo (1993), Daft et al. (1988), Fosfuri and Tribó (2008), Jones et al. (2001), Sidhu et al. (2007)
Acquire 4	The search for relevant information concerning our industry is every-day business in our company.	Daft et al. (1988), Jansen et al. (2005), Wilkens et al. (2004)
Acquire 5	Our management motivates the employees to use information sources within our industry.	Sidhu et al. (2007), Veugelers and Cassiman (1999)
Acquire 6	In our company it is appreciated when employees procure information from other industries as well.	Auster and Choo (1993), Jansen et al. (2005), Veugelers and Cassiman (1999)
Acquire 7	Our management expects that the employees deal with information beyond our industry.	Jansen et al. (2005), Laursen and Salter (2006)
Assimilate 1	In our company ideas and concepts are communicated cross-departmental.	Shu, Wong, and Lee (2005)
Assimilate 2	Our management emphasizes cross-departmental support to solve problems.	Schmidt (2005)
Assimilate 3	Our company uses tools (e.g., intranet, internal studies/reports) to spread knowledge in the whole organization.	Bontis, Crossan, and Hulland (2002)
Assimilate 4	In our company there is a quick information flow, e.g., if a business unit obtains important information it communicates this information promptly to all other business units or departments.	Bontis et al. (2002), Hock-Hai et al. (2006), Tiwana and McLean (2005), Vorhies and Harker (2000)
Assimilate 5	Our management demands periodical cross-departmental meetings to interchange new developments, problems, and achievements.	Farrell (2000), Hult et al. (2004), Kohli et al. (1993), Pavlou and El Sawy (2006), Vorhies and Harker (2000)
Assimilate 6	Our employees of diverse departments get along well, when communicating with each other on a cross-departmental basis.	Ko, Kirsch, and King (2005)
Assimilate 7	For projects our management supports temporary exchange of personnel between departments.	Schmidt (2005)
Assimilate 8	In our company there is informal contact between employees of all levels and departments.	Shu et al. (2005)
Assimilate 9	Our management emphasizes a shared lingo for intra-corporate communication.	Huber (1991), Hult et al. (2004), Ko et al. (2005), Szulanski (1996)
Assimilate 10	In our company employees are conscious about who possesses special skills and knowledge and for who certain information is of interest.	Espinosa et al. (2007), Pavlou and El Sawy (2006), Szulanski (1996)
Assimilate 11	Our employees share their knowledge, their information and their experience willingly with their colleagues.	Gee Woo and Young-Gul (2002), Liao (2006), Liao et al. (2007), Lin (2007), Soonhee and Hyangsoo (2006)
Assimilate 12	Our management is a good role model regarding the distribution of knowledge.	Lu et al. (2006), Szulanski (1996)
Transform 1	Our employees have the ability to structure and use collected knowledge.	Liao et al. (2007)
Transform 2	Our management emphasizes the systematic reuse of insights out of past projects.	Bontis et al. (2002), Hock-Hai et al. (2006)
Transform 3	Our company policy encourages our employees to engage in further training and continuous learning.	Hock-Hai et al. (2006), Nevis and DiBella (1995)

(*continued on next page*)

Table 3 (*continued*)

Item No.	Item	Source (items are based on)
Transform 4	Our employees are used to absorbing new knowledge as well as to prepare it for further purposes and making it available.	Collins and Smith (2006), Jansen et al. (2005), Pavlou and El Sawy (2006)
Transform 5	Our employees successfully link existing knowledge with new insights.	Pavlou and El Sawy (2006)
Transform 6	Our employees cleverly transform information from internal and external sources into valuable knowledge for our company.	Tiwana and McLean (2005)
Transform 7	Our management encourages employees to combine ideas cross-departmentally.	Collins and Smith (2006)
Transform 8	Our management thinks that our learning capabilities are a competitive advantage for our company.	Farrell (2000), Hult et al. (2004), Teo et al. (2006)
Transform 9	Our company owns tools to enhance knowledge that secures the company's competitiveness.	Hock-Hai et al. (2006)
Transform 10	Our employees are able to apply new knowledge in their practical work.	Ettlie and Pavlou (2006)
Transform 11	Our management encourages employees to generate knowledge.	Bontis et al. (2002)
Transform 12	Our management provides employees with enough scope for development to use the aggregated information for experimenting with alternative solution possibilities.	Expert interview
Exploit 1	Our company launches innovative products/services promptly with regard to its research.	Liao (2006)
Exploit 2	Our management supports the development of prototypes.	Nambisan, Agarwal, and Tanniru (1999)
Exploit 3	Our company strives to convert innovative ideas into patents.	Expert interview
Exploit 4	Our company regularly reconsiders technologies and adapts them in accordance with new knowledge.	Expert interview
Exploit 5	Our company has the ability to work more effectively by adopting new technologies.	Expert interview

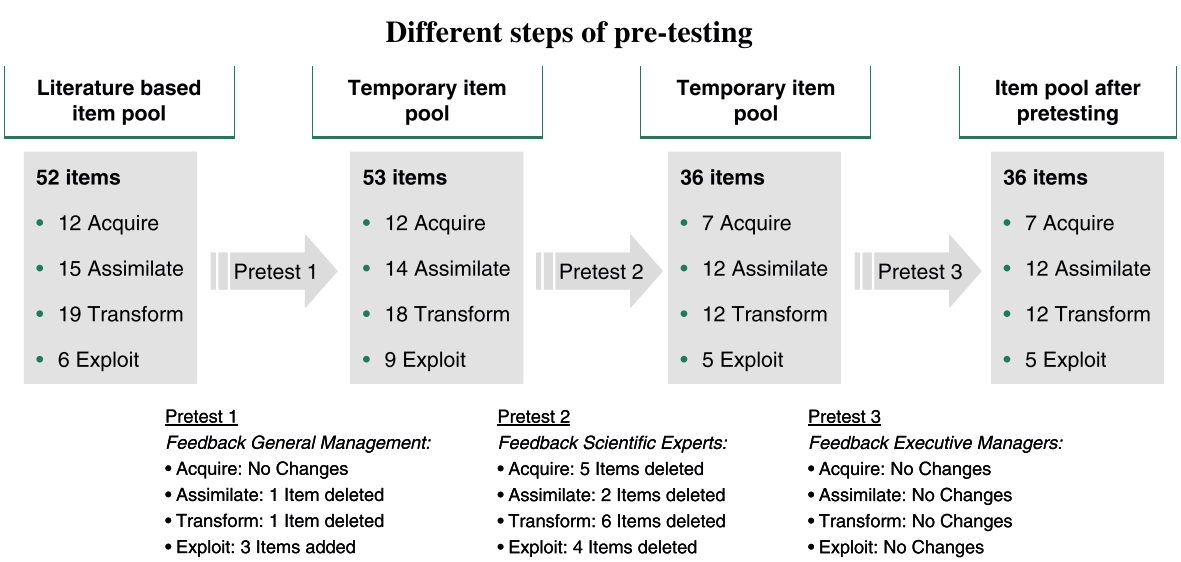

Figure 2 Different steps of pre-testing.

and the explained variance was 19.0%, indicating that common method variance was not a serious problem in the two databases (Reinartz, Krafft, & Hoyer, 2004). Finally, following Newman (2003), completed questionnaires with more than 10% missing values were excluded, and the remaining missing values were estimated by applying the maximization parameter estimation procedure.

Analyses

Analyses progressed in two stages. In the first stage, the first sample tested several theoretically plausible alternative factor structure specifications and eliminated those items that did not adequately capture the theoretical components of the ACAP construct. In the second stage, the second sample replicated the findings obtained from the first sample (Hansen, 2002).

The purpose of the scale development in this study is to develop reflective measures for the four dimensions of the ACAP construct (Jarvis, MacKenzie, Podsakoff, Mick, & Bearden, 2003). Only for these reflective measures are stochastic measures to validate multi-item constructs available. (Compare particularly the approach from DeVellis (2003), which implicitly assumes reflective specifications for the measures to be developed.) Formative measures cannot be developed by means of survey research and stochastic approaches; in any case, there is no established guideline on how to develop these measures in the literature. (Compare, for example, Rossiter, 2002.) For formative measures, qualitative reasoning is more important in determining whether all facets of a construct have been captured.

Study 1: scale refinement

The data for the first sample were analyzed using maximum-likelihood factor analyses with promax rotation. The Kaiser-Meyer-Olkin (MSA) estimate for the data set was .92,

Table 4 Absorptive capacity, factor loadings, and cross loadings (first sample).

Item	Acquisition	Assimilation	Transformation	Exploitation	Remark
Acquire 1	.32	.21	.21	.19	Eliminated due to low factor loading
Acquire 2	.40	.10	.16	.46	Eliminated due to low factor loading
Acquire 3	.39	.17	.20	.31	Eliminated due to low factor loading
Acquire 4	.60	.31	.37	.28	
Acquire 5	.75	.46	.39	.38	
Acquire 6	.85	.40	.34	.29	
Acquire 7	.84	.35	.37	.30	
Assimilate 1	.32	.69	.45	.31	
Assimilate 2	.34	.81	.45	.33	
Assimilate 3	.37	.49	.48	.44	Eliminated due to low factor loading
Assimilate 4	.29	.59	.49	.24	
Assimilate 5	.36	.68	.48	.41	
Assimilate 6	.27	.70	.60	.23	
Assimilate 7	.33	.54	.40	.30	
Assimilate 8	.28	.59	.44	.22	
Assimilate 9	.27	.61	.38	.18	
Assimilate 10	.25	.63	.53	.19	
Assimilate 11	.41	.80	.52	.26	
Assimilate 12	.29	.70	.48	.37	
Transform 1	.44	.56	.77	.37	
Transform 2	.32	.65	.47	.27	Eliminated due to low factor loading
Transform 3	.45	.67	.62	.40	Eliminated due to high cross-loading
Transform 4	.38	.61	.83	.39	
Transform 5	.34	.55	.84	.33	
Transform 6	.42	.56	.86	.44	
Transform 7	.54	.76	.71	.48	Eliminated due to high cross-loading
Transform 8	.38	.56	.60	.38	Eliminated due to high cross-loading
Transform 9	.34	.49	.61	.55	Eliminated due to high cross-loading
Transform 10	.35	.61	.77	.36	
Transform 11	.42	.72	.67	.42	Eliminated due to high cross-loading
Transform 12	.38	.60	.60	.29	Eliminated due to high cross-loading
Exploit 1	.34	.31	.47	.77	
Exploit 2	.27	.30	.28	.77	
Exploit 3	.20	.16	.19	.60	
Exploit 4	.35	.39	.40	.66	
Exploit 5	.30	.35	.36	.64	

indicating that the use of factor analyses was optimally appropriate. The analyses generated four significant factors. Table 4 shows the factor loadings for the first sample, in which the explained variance of the extracted factors was 54.3%, exceeding the standard cut-off point for the explained variance (more than 50%). Five items (Acquire 1, 2, and 3; Assimilate 3; and Transform 2) were eliminated because their factor loadings were less than .5. Another six items (Transform 3, 7, 8, 9, 11, and 12) were eliminated because of cross-loadings, since these items loaded strongly on multiple factors. According to Chin (1998) ''(i)f an indicator loads higher with other (latent variables) than the one it is intended to measure, the researcher may wish to reconsider its appropriateness because it is unclear which construct or constructs it is actually reflecting'' (1998, p. 321). Therefore, following an even more conservative approach than Chin (1998), we eliminated indicators where the factor loading was lower than the cross-loading or less than .1 higher than the cross-loading.

Table 5 Cronbach coefficient alpha for the first sample.

Dimension	Number of items	Cronbach's coefficient α
Acquisition	3	.79
Assimilation	6	.91
Transformation	4	.91
Exploitation	3	.82

To assess the reliability of the scale items and ensure a high Cronbach's coefficient α, items with an item-to-total correlation smaller than .5 or squared multiple correlations (SMC) smaller than .16 (Acquire 6, Assimilate 6, 7, 8, 9, 10, Transform 5, and Exploit 1, 3) were also eliminated (Bearden, Netemeyer, & Teel, 1989). Table 5 displays the Cronbach's coefficient α for the four ACAP dimensions. Following Churchill (1979), coefficient α is the most commonly used measure of internal consistency. As the data in Table 5 indicate, α values ranged from .79 to .91, higher than the standard .7 cut-off point (Nunnally, 1978), supporting the reliability of the four ACAP dimensions.

The four ACAP dimensions, which generated AVEs above the recommended .5 and factor reliabilities exceeding .6, also had acceptable convergent validity (Bagozzi & Yi, 1988). Further, for measures to achieve discriminant validity, their convergent validity coefficients should be greater than or equal to .5 (Fornell & Larcker, 1981) and Table 6 shows that all factors met this criterion.

A review of standardized residual and modification indices identifies potential areas of model misspecification (Saris, Satorra, & Sörbom, 1987). Since there were no unreasonable estimates and all factor loadings were significant, a re-estimation of the model was not required.

Confirmatory factor analysis (CFA) was conducted using AMOS 17.0. All model fit indices (Table 7) were evaluated using multiple criteria that included the Jöreskog and Sörbom's goodness-of-fit index (GFI), the root mean square error of approximation (RMSEA) (Steiger & Lind, 1980), the

Table 6 Fornell—Larcker coefficients for the first sample.

Dimension	Acquisition	Assimilation	Transformation	Exploitation
Acquisition	*.60*			
Assimilation	.24	*.50*		
Transformation	.21	.48	*.60*	
Exploitation	.17	.23	.21	*.50*

Table 7 Summary of the global fit indices for the first sample.

Model dimension	Content	GFI	RMSEA	χ^2/df	AGFI	NFI	CFI	SRMR
One	All items	.68	.19	13.37	.57	.64	.66	.12
Two	Identify, assimilate plus exploit (first order uncorrelated)	.83	.14	6.20	.74	.81	.84	.10
Two	Acquire, assimilate plus transform, exploit (second order uncorrelated)	.76	.16	10.64	.67	.69	.71	.14
Two	PACAP (acquire, assimilate) plus RACAP (transform, exploit) (third order uncorrelated)	.93	.07	3.14	.89	.91	.94	.11
Three	First order: identify, assimilate, exploit	.84	.09	3.47	.79	.83	.87	.06
Three	Second order: identify, assimilate, exploit (uncorrelated)	.84	.93	3.47	.79	.83	.87	.05
Three	Second order: identify, assimilate, exploit (correlated)	.83	.10	3.78	.77	.81	.85	.10
Four	First order: acquisition, assimilate, transform, exploit	.92	.06	1.85	.89	.91	.96	.05
Four	Second order: PACAP: acquisition, assimilate, RACAP: transform, exploit (uncorrelated)	.92	.05	1.84	.89	.91	.96	.05
Four	Second order: PACAP: acquisition, assimilate, RACAP: transform, exploit (correlated)	.91	.07	2.28	.87	.89	.93	.12

χ^2/degrees of freedom ratio (Wheaton, Muthen, Alwin, & Summers, 1977), the AGFI, Bentler and Bonett's (1980) NFI, the comparative fit index (CFI) (Hu & Bentler, 1995), and the standardized root mean square residual (SRMR). The results for all models, which were developed based on the competing theories, fit the remaining 16 items shown in Table 7. Corresponding to these results, an uncorrelated second-order four-dimension ACAP model was chosen for subsequent analyses (GFI = .92, RMSEA = .05, χ^2/df = 1.84, AGFI = .89, NFI = .91, CFI = .96, SRMR = .05). Compared to all other estimated models, this model shows the best fit indices, reaffirming its superiority, especially over the Cohen and Levinthal (1990) three-dimension conceptualization of ACAP (Baumgartner & Homburg, 1996; Bentler & Bonett, 1980). Table 7 displays all tested models, which differ in the numbers of factors, vertical depths (first-order, second-order and third-order constructs), and correlations.

Table 8 Cronbach coefficient alpha for the second sample.

Dimension	Number of items	Cronbach coefficient α
Acquisition	3	.73
Assimilation	4	.85
Transformation	4	.93
Exploitation	3	.80

Study 2: scale validation

Upon completing scale refinement, the second sample replicated the initial findings for validation of the ACAP scale. Table 8 shows the Cronbach's coefficient α for all four

Table 9 Fornell—Larcker coefficients for the second sample.

Dimension	Acquisition	Assimilation	Transformation	Exploitation
Acquisition	*.52*			
Assimilation	.24	*.57*		
Transformation	.21	.41	*.76*	
Exploitation	.17	.24	.21	*.61*

Table 10 Final ACAP scale.

Final ACAP scale	
Acquisition	
Please specify to what extent your company uses external resources to obtain information (e.g., personal networks, consultants, seminars, internet, database, professional journals, academic publications, market research, regulations, and laws concerning environment/technique/health/security):	
Acquire 4	The search for relevant information concerning our industry is every-day business in our company.
Acquire 5	Our management motivates the employees to use information sources within our industry.
Acquire 7	Our management expects that the employees deal with information beyond our industry.
Assimilation	
Please rate to what extent the following statements fit the communication structure in your company:	
Assimilate 1	In our company ideas and concepts are communicated cross-departmental.
Assimilate 2	Our management emphasizes cross-departmental support to solve problems.
Assimilate 4	In our company there is a quick information flow, e.g., if a business unit obtains important information it communicates this information promptly to all other business units or departments.
Assimilate 5	Our management demands periodical cross-departmental meetings to interchange new developments, problems, and achievements.
Transformation	
Please specify to what extent the following statements fit the knowledge processing in your company:	
Transform 1	Our employees have the ability to structure and to use collected knowledge.
Transform 4	Our employees are used to absorb new knowledge as well as to prepare it for further purposes and to make it available.
Transform 6	Our employees successfully link existing knowledge with new insights.
Transform 10	Our employees are able to apply new knowledge in their practical work.
Exploitation	
Please specify to what extent the following statements fit the commercial exploitation of new knowledge in your company (NB: Please think about all company divisions such as R&D, production, marketing, and accounting):	
Exploit 2	Our management supports the development of prototypes.
Exploit 4	Our company regularly reconsiders technologies and adapts them accordant to new knowledge.
Exploit 5	Our company has the ability to work more effective by adopting new technologies.

Table 11 Post-hoc analyses for sub-samples.

	Age		Size		Customer focus		Performance	
	Young companies	Established companies	Small companies	Large companies	B2B	B2C	More successful	Less successful
GFI	.92	.91	.92	.81	.93	.85	.86	.91
AGFI	.88	.87	.90	.73	.90	.80	.80	.87
CFI	.97	.95	.95	.94	.97	.93	.94	.94
NFI	.91	.90	.92	.78	.93	.83	.87	.90
TLI	.96	.94	.94	.93	.96	.91	.93	.93
RMSEA	.05	.07	.07	.07	.06	.08	.08	.07
SRMR	.05	.06	.05	.10	.05	.06	.06	.06
χ^2/df	1.39	1.82	2.30	1.30	1.70	1.60	1.67	2.20

dimensions. All values exceed the recommended .7 cut-off point (Nunnally, 1978).

Acceptable convergent validity of the second sample was also supported (Table 9), with all four dimensions having AVEs exceeding .5 and factor reliabilities larger than .6 (Bagozzi & Yi, 1988). The convergent validity coefficients were greater than or equal to .5, underscoring the discriminant validity of each of the four ACAP factor measures. This information appears in Table 9.

The standardized residual and modification indices for identifying potential areas of model misspecification were also reviewed (Saris et al., 1987), and revision of the modified indices resulted in the elimination of two items (Assimilate 11 and 12) in order to achieve a better model fit for the ACAP construct (Jöreskog, 1993). The model fit for the ACAP construct met established criteria (GFI = .94, RMSEA = .06, χ^2/df = 2.18, AGFI = .91, NFI = .95, CFI = .97, SRMR = .05). Table 10 displays the validated scale items.

As stated above, the purpose of the present study was to develop reflective measures, and we did so following DeVellis (2003), who provides guidelines for developing these reflective measures. To substantiate the reflective specification further, we verified the specification *ex post* by means of qualitative criteria and stochastic tests. Jarvis et al. (2003) state that reflective constructs, unlike formative measures, are characterized by a causality which runs from the construct to the items, an interchangability of items, a expected high correlation between items, and items that share a similar nomological net. We applied these four criteria to our four ACAP dimensions and concluded that our constructs largely fulfilled these criteria, which finding substantiates the reflective specification. Further, Bollen and Ting (2000) propose a tetrad test, which allows a specification to be verified by means of survey data. We conducted a tetrad test for all four dimensions of the ACAP construct, testing the null hypothesis that constructs have a reflective specification,[1] and found that reflective specifications are suitable for our constructs.

[1] The tetrad test can only be applied to constructs with four or more items. In order to conduct this test for our constructs ''acquisition'' and ''exploitation'' which only embrace three items, a forth item was randomly added to conduct this test. This approach is in line with Bollen and Ting (2000).

Further analyses

In order to obtain a generalizable scale, the present study builds upon random samples that incorporate a broad set of companies (e.g., in terms of company age) in selected research-intensive sectors. Since knowledge processes may vary between companies of different ages or other characteristics (McAdam & Reid, 2001), we conducted some post hoc investigations. First, we divided our second sample into young and established companies and conducted confirmatory factor analyses of both groups. Table 11 shows that the developed scale provides satisfactory psychometric properties for both sub-samples, indicating that the scale is appropriate for both young and established companies. The same procedure applied for subgroups of smaller and larger companies. We also tested a potential bias caused by the customer focus of the companies (B2B vs. B2C), and Table 11 indicates that both subsamples revealed satisfactory psychometric properties of the scale. Further, our sampling procedure could imply a bias if more successful companies (e.g., companies with superior knowledge processes) were more likely to take part in the survey. In order to exclude the possibility of this bias, we split the sample into two sub-samples of the same size based on firm performance (measured by growth compared to the prior year and average growth since founding). Again, our scale shows satisfactory psychometric properties for both sub-samples, indicating that the scale development was not significantly influenced by this bias.

Discussion

Theoretical implications

The acknowledged role of ACAP in knowledge sharing, organizational learning and capability building underscores the need to understand its various dimensions (Lane et al., 2006; Zahra & George, 2002). Researchers have conceptualized and measured ACAP differently, making it difficult to compare prior findings and establish their theoretical and managerial relevance. The development and presentation of a four-factor measure of ACAP helps to ensure valid

results and facilitates comparisons across studies. The proposed measure assesses the degree to which a company engages in knowledge acquisition activities, assimilates acquired information into existing knowledge, transforms the newly adapted knowledge, and commercially exploits the transformed knowledge to its competitive advantage.

This study addresses a debate in the literature on the appropriate measurement of ACAP. Specifically, by developing 14 items and classifying them into four dimensions, this study helps researchers conduct more systematic analyses. The proposed measures can also serve as a foundation from which to compare findings across studies and research settings, making it possible to apprehend the value-added of ACAP as a construct. For example, the results show that transformation is an integral dimension of ACAP, contrary to Torodova and Durisin's (2007) position. Consequently, recognizing transformation when measuring ACAP, ''helps to open the black box that has dominated the prior research'' (Zahra & George, 2002, p. 190) by highlighting the firm's ability to modify its knowledge for use (Kim, 1997) and gain value from investments made in developing ACAP. The other three dimensions of ACAP do not capture transformation, reinforcing the distinctiveness of transformation as a separate dimension, so the four-dimension definition of ACAP is strongly supported empirically.

Managerial implications

This study provides managers a useful tool with which to assess their companies' strengths and weaknesses in regard to ACAP. The proposed measures make it possible to compare a firm's ACAP to those of other firms, providing a basis for determining where additional investments should be made to upgrade and improve the use of ACAP. Managers can creatively leverage their firms' ACAP by conceiving of and exploring ways to integrate the four ACAP dimensions. Strategic heterogeneity, a major source of competitive advantage, results from managers' efforts to configure their firms' ACAP and from managers' resourcefulness and ingenuity in using ACAP to create varied and new products, systems and processes that distinguish the company from its rivals. Like other intangible resources, ACAP requires managerial attention and sustained investment.

Limitations and opportunities for further research

A key limitation of this study is its focus on research-intensive firms in Germany. Companies in these sectors face challenges in acquiring and processing knowledge from outside sources and keeping their ACAP current, but the question remains concerning whether our scale holds in different contexts. Without further evidence, one cannot conclude that our scale applies in the same manner to other industries or countries, so future research should conduct scale developments for the ACAP constructs in, for example, other countries that are characterized by certain national cultures and stages of macroeconomic development (Hofstede, 2001). For this purpose, initial pools of items that reflect the particularities of these contexts need to be derived and, before the quantitative analysis, extensive qualitative pre-tests, such as team discussion or interviews with practitioners, should be conducted in order to understand the activities that form the ACAP dimensions in these specific contexts and should, therefore, be part of a reliable and valid scale (Douglas & Craig, 1984). When different scales are developed in specific contexts, one can compare these scales with our study to detect the differences and similarities between scales.

A related issue that requires additional empirical research is the relative importance of the four ACAP dimensions in different settings. Does this importance vary along the stages of the organizational life cycle? Do they change as the industry evolves? Do they vary between knowledge-intensive industries and other kinds of industry? It is also important to determine the relative importance that the four different dimensions play in determining various organizational outcomes (e.g., technological capabilities, success in alliances, and organizational performance).

Building upon our suggestions for the ACAP measure, researchers could also probe how companies go about building the routines (and the knowledge underlying them) that make up each of the four dimensions. Do younger and established companies go about accomplishing building these routines differently? How do these companies use their social and relational capital to gain the knowledge that is essential to building each dimension? Do they use different coordination techniques and integrative mechanisms? If so, what are the strategic consequences of these differences for a company's performance? Addressing these issues can enrich ongoing discussions of the role of intangible resources and capabilities in creating competitive advantage (Barney, 1991; Helfat, 1997; Penrose, 1959) and discussions of the role of knowledge in building, replenishing, and upgrading organizational capabilities (Mowery, Oxley, & Silverman, 1996; Teece et al., 1997). Appreciating how different ACAP dimensions may vary across time and settings could also enrich discussions on ACAP's role in creating the new knowledge that positions firms to evolve and grow (Nonaka, 1994). Such an understanding can add to the young but growing literature that uses the knowledge-based theory of the firm.

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TESSA C. FLATTEN is a post doctoral candidate at the chair for Business Administration and Sciences for Engineers and Scientists at RWTH Aachen University, Germany. She received her master's degree in business administration from RWTH Aachen University, Germany. Her research focuses innovation management in particular absorptive capacity. She has presented her research at leading international conferences as the AoM annual meeting.

ANDREAS ENGELEN is assistant professor at the chair for Business Administration and Sciences for Engineers and Scientists at RWTH Aachen University, Germany. He received his doctoral degree in business administration from RWTH Aachen University, Germany. His areas of research interest include international marketing, international management and entrepreneurial marketing. He has presented his research at leading international marketing and entrepreneurship conferences and has published in prominent marketing journals such as *Journal of International Marketing* and the *Entrepreneurship: Theory and Practice*.

SHAKER A. ZAHRA holds the Robert E. Buuck Chair of Entrepreneurship and is Professor of Strategy and Organization in the Carlson School of Management at the University of Minnesota, where he is the Director of the Gary S. Holmes Center for Entrepreneurship. His research focuses on corporate and international entrepreneurship in high technology, global companies and industries. His articles have appeared in the Academy of Management Journal, Academy of Management Review, Strategic Management Journal, Journal of Management, Journal of Management Studies, Research Policy, Industrial and Corporate Change, Information Systems Research, Decision Sciences, European Management Journal, among others. His research, teaching and professional service have received several awards.

MALTE BRETTEL is University Professor for Business Administration and Sciences for Engineers and Scientists at RWTH Aachen University, Germany. He received his doctoral degree and his postdoctoral qualification from WHU Otto Beisheim School of Management. He is co-founder of JustBooks (today ABEBooks). His areas of research interest include entrepreneurial management and development, entrepreneurial marketing, entrepreneurial finance and innovation management. He has published his work in various books and journals and has presented his research at leading international conferences, including the AMA Summer Marketing Conference, the AOM Annual Meeting, and the Babson Entrepreneurship Conference.